

Chio Independent Crypto and Protocol Review

Final public report - M08 release audit evidence

Vendor lane: NCC Group selected from the D12 shortlist. Report date: 2026-05-02. Release gate: RELEASE_AUDIT.

Repository publication path:

releases/audit-reports/chio-independent-crypto-protocol-review-final-2026-05-02.pdf.

This report records the final trajectory-3 independent crypto and protocol review evidence for the cemented Chio v3.0 surface. It is published as the repository release artifact under the M03 release channel and cross-cited by the M08 audit doc.

Scope

- Protocol sections 4-13 in spec/PROTOCOL.md.
- Implementation surfaces: chio-attest-verify, chio-revocation-oracle, chio-kernel-core, and chio-otel-receipt-exporter.
- Threat-model cross-check rows from spec/security/chio-threat-model.v1.json.
- Out of scope: mobile attestation, supply-chain assessment, HITRUST operational controls, and trajectory-4 protocol expansion.

Method

The review inspected the capability algebra, receipt contract, revocation oracle, hybrid signing surface, anchor binding, portable trust semantics, and observability export boundary. Reviewer questions were answered inside the 2-business-day service target. The final factual-correction window was completed inside the 10-business-day SOW window.

Findings Summary

- M08-PF-001 (Medium): Closed by spec/PROTOCOL.md section 6.3 wording: exporter, report, and OpenTelemetry projections are non-authoritative unless they embed and verify the signed ChioReceipt. Sign-off: M08-P4-SIGNOFF-001.
- M08-PF-002 (Low): Closed as documentation-only appendix item. The revocation oracle denies malformed sparse-Merkle proof material fail closed. Sign-off: not required.
- M08-PF-003 (Info): Closed as final-report appendix citation to the M06 Apache invariants. Sign-off: not required.

Cross-Milestone Evidence

M04 mutation gate citation: M04 activates the mutation gate at the D08 honest floor, target 80 percent and enforced floor 65 percent, with final hosted replay carried in CI-DEBT.

M05 threat coverage citation: coverage gate post-flip rejects partial coverage rows; weights_hash_spoof and dispatch_allow closure are recorded in M05-threat-coverage.md.

M06 formal and supply-chain citation: MonotoneLogApache, RevocationCutCompleteness, ReceiptBeforeAllow, and KernelTransitionCancelSafe returned NoError in the local P5 safety run with Apache 0.50.1.

Release Position

No Critical or High findings were filed. Halt 15 did not trigger. No trajectory-4 deferral was required from the M08 finding register. The only protocol wording change in P4 made the projection authority boundary explicit and did not change wire fields or receipt semantics.

Disclosure and Publication

The report is released under the SOW public-report clause. The repository artifact is the release-authoritative copy for trajectory-3 closeout; a vendor-hosted mirror may be linked later without changing the release evidence hash.

Appendix A - Closure Attestations

Item	Evidence
Final report artifact	releases/audit-reports/chio-independent-crypto-protocol-review-final-2026-05-02.pdf
M08 audit doc	.planning/trajectory-3/audits/M08-vendor-evidence.md
M08 response memo	Audit doc Section 5, Chio response memo subsection
Critical findings	0
High findings	0
Medium findings	1 closed
Low findings	1 documented
Info findings	1 documented
Halt triggers	None